

SGN – Assignment #1

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1 Periodic orbit

Exercise 1

Consider the 3D Earth–Moon Circular Restricted Three-Body Problem with $\mu = 0.012150$. Note that the CRTBP has an integral of motion, that is, the Jacobi constant

$$J(x, y, z, v_x, v_y, v_z) := 2\Omega(x, y, z) - v^2 = C$$

where $\Omega(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}\mu(1-\mu)$ and $v^2 = v_x^2 + v_y^2 + v_z^2$.

- 1) Find the coordinates of the five Lagrange points L_i in the rotating, adimensional reference frame with at least 10-digit accuracy and report their Jacobi constant C_i .

Solutions to the 3D CRTBP satisfy the symmetry

$$\mathcal{S} : (x, y, z, \dot{x}, \dot{y}, \dot{z}, t) \rightarrow (x, -y, z, -\dot{x}, \dot{y}, -\dot{z}, -t).$$

Thus, a trajectory that crosses perpendicularly the $y = 0$ plane twice is a periodic orbit.

- 2) Given the initial guess $\mathbf{x}_0 = (x_0, y_0, z_0, v_{x0}, v_{y0}, v_{z0})$, with

$$\begin{aligned} x_0 &= 1.068792441776 \\ y_0 &= 0 \\ z_0 &= 0.071093328515 \\ v_{x0} &= 0 \\ v_{y0} &= 0.319422926485 \\ v_{z0} &= 0 \end{aligned}$$

Find the periodic halo orbit having a Jacobi Constant $C = 3.09$; that is, develop the theoretical framework and implement a differential correction scheme that uses the STM, either approximated through finite differences **or** achieved by integrating the variational equation.

Hint: Consider working on $\varphi(\mathbf{x} + \Delta\mathbf{x}, t + \Delta t)$ and $J(\mathbf{x} + \Delta\mathbf{x})$ and then enforce perpendicular cross of $y = 0$ and Jacobi energy.

The periodic orbits in the CRTBP exist in families. These can be computed by ‘continuing’ the orbits along one coordinate or one parameter, e.g., the Jacobi energy C . The *numerical continuation* is an iterative process in which the desired variable is *gradually* varied, while the rest of the initial guess is taken from the solution of the previous iteration, thus aiding the convergence process.

- 3) By gradually decreasing C and using numerical continuation, compute the families of halo orbits until $C = 3.04$.

(8 points)

1 Periodic orbit

1.1 Lagrange Points and Jacobi Constants

Methodology The Lagrange points represent the equilibrium solutions of the Circular Restricted Three-Body Problem (CRTBP). These points are determined by solving the equation:

$$\frac{\partial \Omega}{\partial x} = \frac{\partial \Omega}{\partial y} = \frac{\partial \Omega}{\partial z} = 0 \quad (1)$$

where Ω is the potential function. The equations are solved numerically using the MATLAB function `fsolve`, with the initial guesses derived from theoretical considerations. Specifically, the z -component must be zero for all libration points to guarantee equilibrium. For the collinear points (L_1 , L_2 , L_3), the y -component is also zero. Finally, for the triangular points L_4 and L_5 , the initial guess is derived based on simple geometrical considerations.

The solver was configured with an **Optimality Tolerance** of 10^{-12} to ensure 10-digit accuracy. Finally, the Jacobi constants for each point were computed.

Results As expected, the Lagrange points of the Earth-Moon CRTBP lie in the plane passing through Earth and the Moon. The coordinates and Jacobi constants of the Lagrange points are listed in Table 1. The spatial distribution of these points is illustrated in Fig. 1.

	L_1	L_2	L_3	L_4	L_5
x	0.8369180073	1.1556799131	-1.0050624018	0.4878500000	0.4878500000
y	0.0000000000	0.0000000000	0.0000000000	0.8660254038	-0.8660254038
C	3.2003380950	3.1841582164	3.0241489429	3.0000000000	3.0000000000

Table 1: Lagrangian points coordinates and Jacobi constants.

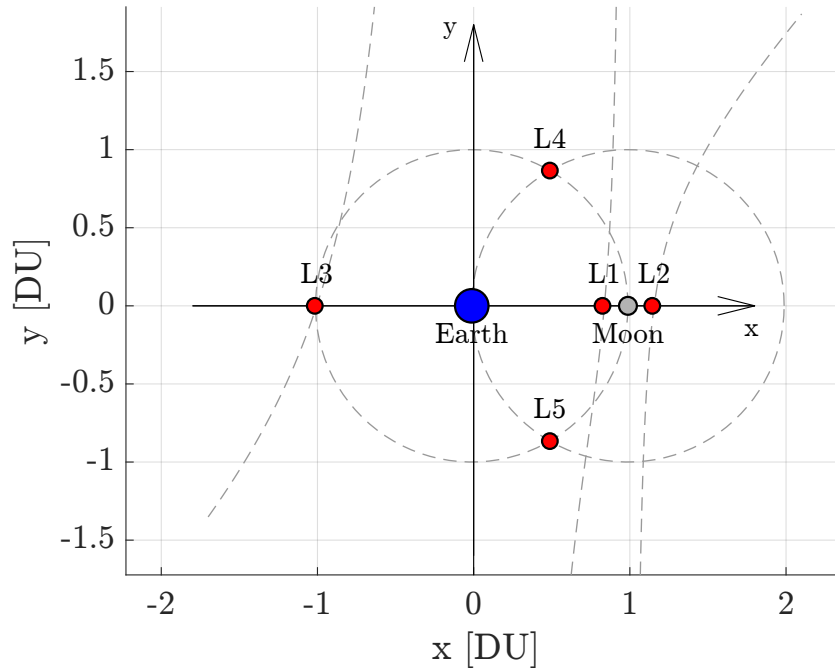


Figure 1: Lagrange points in the Earth-Moon CRTBP (@EMB Earth-Moon rotating frame).

1.2 Periodic Halo Orbit

Methodology To compute a periodic halo orbit with a Jacobi constant $C = 3.09$, the provided initial guess is corrected using a differential correction scheme. First, the CRTBP dynamics are modeled and integrated in time together with the State Transition Matrix (STM). The STM is computed using the variational approach. The complete set of equations of motion is as follows:

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = \frac{\partial \Omega}{\partial x} + 2v_y \\ \dot{v}_y = \frac{\partial \Omega}{\partial y} - 2v_x \\ \dot{v}_z = \frac{\partial \Omega}{\partial z} \end{cases} \quad \begin{cases} \dot{\Phi}(t_0, t) = \mathbf{A}(t)\Phi(t_0, t) \\ \Phi_0 = \mathbf{I}_{6 \times 6} \end{cases} \quad (2)$$

where Φ is the STM, Ω is the potential, and $\mathbf{A}(t) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \Big|_{\mathbf{x}=\mathbf{x}^*}$, with $\mathbf{x}^*$ as reference trajectory.

The goal is to correct the initial state to achieve the periodicity and obtain the desired halo orbit ($C = 3.09$).

The implemented correction scheme is based on a first-order Taylor expansion of the flow with respect to the initial state and final time:

$$\varphi(\mathbf{x}_0 + \Delta \mathbf{x}, t + \Delta t) = \varphi(\mathbf{x}_0, t) + \frac{\partial \varphi}{\partial \mathbf{x}_0} \Delta \mathbf{x} + \frac{\partial \varphi}{\partial t} \Delta t. \quad (3)$$

It should be noted that $\frac{\partial \varphi}{\partial \mathbf{x}_0}$ represents the State Transition Matrix (STM) $\Phi(t_0, t_f)$ at the final time. Similarly, the Jacobi constant is linearized around the initial state:

$$J(\mathbf{x}_0 + \Delta \mathbf{x}) = J(\mathbf{x}_0) + \frac{\partial J}{\partial \mathbf{x}_0} \Delta \mathbf{x}. \quad (4)$$

To develop the theoretical framework for the corrections, an augmented 7×7 linear system is constructed, as shown in Eq. (5). The left-hand side is represented by the error vector $\mathbf{e}$, which contains the differences between the targeted state and its computed values. The right-hand side is composed by the augmented matrix $\mathbf{B}$ multiplied by the vector of corrections $\Delta \mathbf{x}$. The matrix $\mathbf{B}$ is composed by the STM at the final time (6×6), the derivative of the state vector at the final time, $\dot{\mathbf{x}}_f$ (7×1), and the gradient of the Jacobi constant (1×7) with respect to the initial state $\mathbf{x}_0$, as presented in Eq. (4).

The correction vector $\Delta \mathbf{x}$ includes Δx_0 , appended with the time correction Δt :

$$\mathbf{e} = \mathbf{B} \Delta \mathbf{x}, \quad \mathbf{B} = \begin{bmatrix} \Phi(t_0, t_f) & \dot{\mathbf{x}}_f \\ \nabla J(\mathbf{x}_0) & 0 \end{bmatrix}_{7 \times 7}. \quad (5)$$

Thanks to the symmetry of the CRTBP, to achieve periodicity it suffices to constrain y , v_x , and v_z to be zero at the final time. Moreover, J must remain constant over time and equal to the target value of $C = 3.09$. The system is thus reduced by considering only the corresponding rows of the matrix $\mathbf{B}$. In addition, the initial state must be corrected only in terms of x_0 , z_0 , v_{y0} , and t_f . This consideration leads to a reduction of the problem to a 4×4 system.

The adopted correction scheme is presented below:

$$\begin{bmatrix} x_0 \\ z_0 \\ v_{y0} \\ t_f \end{bmatrix}_{k+1} = \begin{bmatrix} x_0 \\ z_0 \\ v_{y0} \\ t_f \end{bmatrix}_k + \begin{bmatrix} \Phi_{21} & \Phi_{23} & \Phi_{25} & \dot{y}(t_f) \\ \Phi_{41} & \Phi_{43} & \Phi_{45} & \dot{v}_x(t_f) \\ \Phi_{61} & \Phi_{63} & \Phi_{65} & \dot{v}_z(t_f) \\ \frac{\partial J}{\partial x_0} & \frac{\partial J}{\partial z_0} & \frac{\partial J}{\partial v_{y0}} & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 - y_f \\ 0 - v_{x,f} \\ 0 - v_{z,f} \\ C_{\text{target}} - C \end{bmatrix}_k. \quad (6)$$

The reduced linear system is solved iteratively using a pseudo-Newton method, updating the initial state and final time until the errors on y , v_x , v_z , and C fall below a predefined threshold of 10^{-12} .

Finally, the solution found is propagated forward and backward until the xz -plane crossing time to obtain the halo orbit.

Results The corrected initial state for the halo orbit with $C = 3.09$ is presented in Table 2. In Fig. 2, the corresponding halo orbit around L_2 is shown.

x	y	z
1.0590402077	0.0000000000	0.0739277378
v_x	v_y	v_z
0.0000000000	0.3469245709	0.0000000000

Table 2: Corrected initial state of the halo orbit with $C = 3.09$.

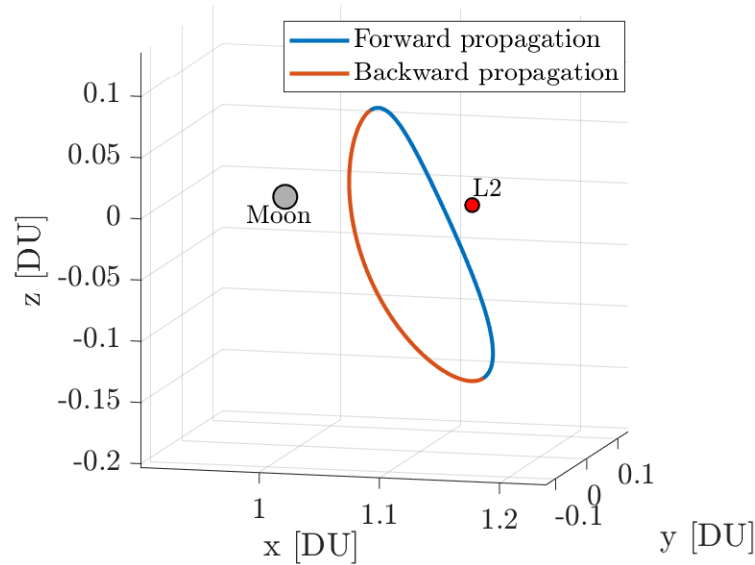


Figure 2: Halo orbit $C = 3.09$ in Earth-Moon CRTBP (@EMB Earth-Moon rotating frame).

1.3 Numerical Continuation for Halo Orbit Families

Methodology The target halo orbit is calculated using the numerical continuation, which involves progressively lowering the Jacobi constant. Starting with $C = 3.09$ and gradually falling to $C = 3.04$, a linearly spaced vector of Jacobi constant values (C) is generated.

For each step in the continuation, the procedure outlined in Section 1.2 is repeated iteratively. The corrected initial state, obtained at each step, is used as initial guess for the subsequent iteration.

To visualize the evolution of each calculated halo orbit, they are propagated forward and backward in time and compared graphically.

Results Table 3 presents the corrected initial state for the halo orbit with $C = 3.04$.

As can be seen in Fig. 3, as the Jacobi constant C falls, the halo orbits' size and distance from L_2 both increase. Since L_2 is linked to a Jacobi constant of approximately $C = 3.184$, this behavior is in line with theoretical predictions.

x	y	z
1.0125655235	0.0000000000	0.0672339583
v_x	v_y	v_z
0.0000000000	0.5103251959	0.0000000000

Table 3: Corrected initial state of the halo orbit with $C = 3.04$.

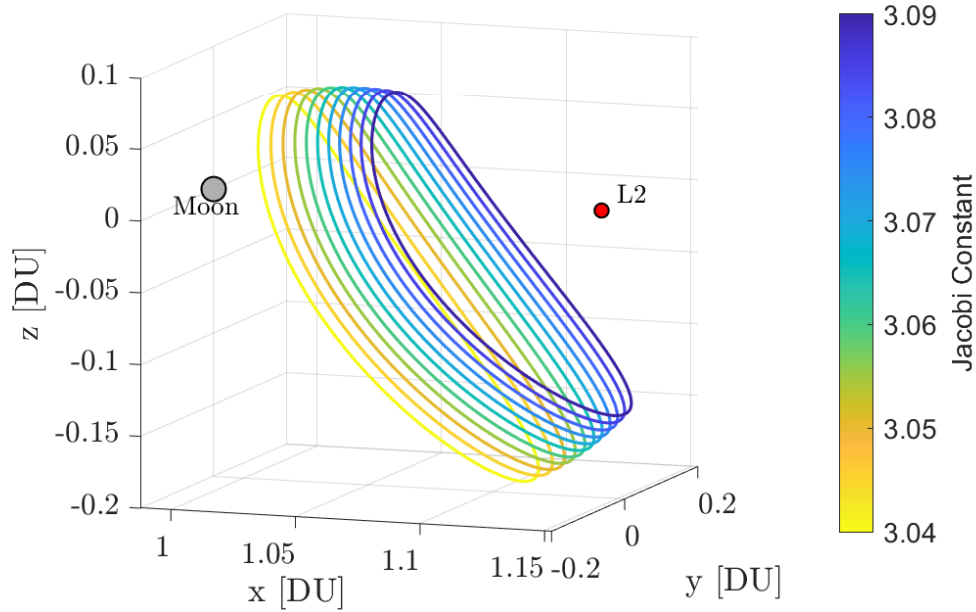


Figure 3: Halo orbits family from $C = 3.09$ to $C = 3.04$ in the Earth-Moon CRTBP (*@EMB Earth-Moon rotating frame*).

2 Impulsive guidance

Exercise 2

Consider the two-impulse transfer problem stated in Section 3.1 (Topputo, 2013)*.

- 1) Using the procedure in Section 3.2, produce a first guess solution using $\alpha = 0.2\pi$, $\beta = 1.41$, $\delta = 4$, and $t_i = 2$. Plot the solution in both the rotating frame and Earth-centered inertial frame (see Appendix 1 in (Topputo, 2013)). Consider the parameters listed in Table 4 and extract the radius and gravitational parameters of the Earth and Moon from the provided kernels and use the latter to compute the parameter μ .

Symbol	Value	Units	Meaning
m_s	3.28900541×10^5	-	Scaled mass of the Sun
ρ	3.88811143×10^2	-	Scaled Sun-(Earth+Moon) distance
ω_s	$-9.25195985 \times 10^{-1}$	-	Scaled angular velocity of the Sun
ω_{em}	$2.66186135 \times 10^{-1}$	s^{-1}	Earth-Moon angular velocity
l_{em}	3.84405×10^8	m	Earth-Moon distance
h_i	167	km	Altitude of departure orbit
h_f	100	km	Altitude of arrival orbit
DU	3.84405000×10^5	km	Distance Unit
TU	4.34256461	days	Time Unit
VU	1.02454018	km/s	Velocity Unit

Table 4: Constants to be considered to solve the PBRFBP. The units of distance, time, and velocity are used to map scaled quantities into physical units.

- 2) Considering the first guess in 1) and using $\{\mathbf{x}_i, t_i, t_f\}$ as variables, solve the problem in Section 3.1 with simple shooting in the following cases
 - a) without providing any derivative to the solver, and
 - b) by providing the derivatives and by estimating the state transition matrix with variational equations.
- 3) Considering the first guess solution in 1) and the procedure in Section 3.3, solve the problem with multiple shooting taking $N = 4$ and using the variational equation to compute the Jacobian of the nonlinear equality constraints.
- 4) Perform an n-body propagation using the solution $\{\mathbf{x}_i, t_i, t_f\}$ obtained in point 2), transformed in Earth-centered inertial frame and into physical units. To move from 2-D to 3-D, assume that the position and velocity components in inertial frame are $r_z(t_i) = 0$ and $v_z(t_i) = 0$. To perform the propagation it is necessary to identify the epoch t_i . This can be done by mapping the relative position of the Earth, Moon and Sun in the PCRTBP to a similar condition in the real world:
 - a) Consider the definition of $\theta(t)$ provided in Section 2.2 to compute the angle $\theta_i = \theta(t_i)$. Note that this angle corresponds to the angle between the rotating frame x -axis, aligned to the position vector from the Earth-Moon System Barycenter (EMB) to the Moon, and the Sun direction.
 - b) The angle θ ranges between $[0, 2\pi]$ and it covers this domain in approximately the revolution period of the Moon around the Earth.

*F. Topputo, “On optimal two-impulse Earth–Moon transfers in a four-body model”, *Celestial Mechanics and Dynamical Astronomy*, Vol. 117, pp. 279–313, 2013, DOI: 10.1007/s10569-013-9513-8.

- c) Solve a zero-finding problem to determine the epoch at which the angle Moon-EMB-Sun is equal to θ_i , considering as starting epoch 2024 Sep 28 00:00:00.000 TDB.
Hints: Exploit the SPK kernels to define the orientation of the rotating frame axes in the inertial frame for an epoch t . Consider only the projection of the EMB-Sun position vector onto the so-defined x-y plane to compute the angle (planar motion).

Plot the propagated orbit and compare it to the previously found solutions.

(11 points)

2 Impulsive guidance

2.1 First Guess Generation

Methodology To produce a first guess solution, the procedure described in Section 3.2 of the referenced paper (Topputo, 2013) is employed. The specified parameters $\alpha = 0.2\pi$, $\beta = 1.41$, $\delta = 4$, and $t_i = 2$ are used to construct the transfer trajectory in the rotating frame.

The Earth and Moon's gravitational parameters and radii are extracted from the provided kernels. The parameter μ , representing the Earth-Moon system's mass ratio in the Planar Bicircular Restricted Four-Body Problem (PBRFBP), is computed as:

$$\mu = \frac{GM_{\text{Moon}}}{GM_{\text{Earth}} + GM_{\text{Moon}}} \quad (7)$$

where GM_{Earth} and GM_{Moon} are Earth and Moon gravitational parameters.

The initial state is then propagated according to the dynamics of the PBRFBP (Topputo, 2013, Section 2.2) in the Earth-Moon barycentric rotating reference frame (EM-barycenter RF). The resulting solution is then rotated into the Earth-Centered Inertial (ECI). See Appendix D for the procedure.

Results From the computations, the value of $\mu = 0.012150584269940$ was retrieved, and the initial guess for the next steps was determined. Fig. 4 shows the trajectory in the Earth-Moon rotating frame. As can be observed, the final propagated state does not reach the Moon parking orbit, implying that the first guess solution fails to provide an acceptable transfer. As a result, the initial state shown in Table 5 solely serves as a preliminary estimate for further optimization processes. The same result can be observed in the Earth-Centered Inertial (ECI) frame, as shown in Fig. 5.

$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]
0.001624	0.010008	-6.302738	8.674975

Table 5: Initial guess in the Earth-Moon rotating frame.

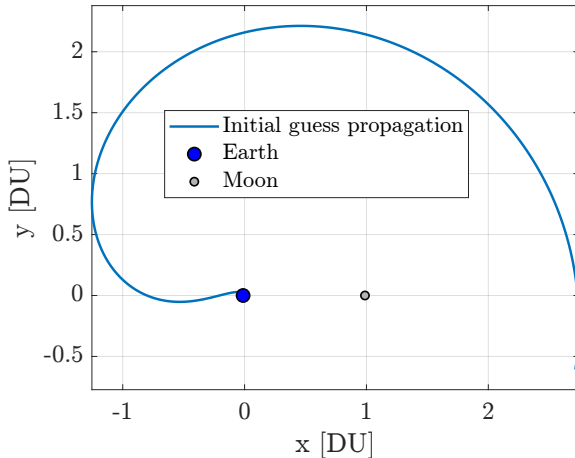


Figure 4: Transfer trajectory (@EMB Earth-Moon rotating frame).

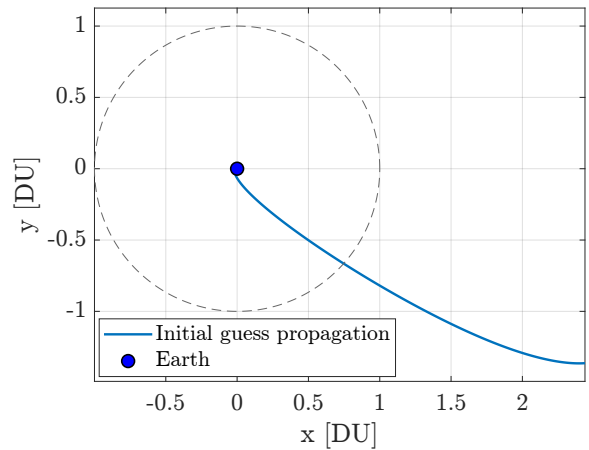


Figure 5: Transfer trajectory (@Earth ECI).

2.2 Two-Impulse Transfer Optimization with Simple Shooting

Methodology In this section, the Earth-Moon transfer is optimized using a simple shooting technique. To clearly outline the procedure to achieve the solutions, the problem statement is

briefly presented as follows:

Let $\mathbf{x}_i$ be the initial state at the Earth circular parking orbit (of adimensionalized radius r_i), t_i and t_f the initial and final adimensionalized propagation times, respectively.

Let $\mathbf{x}_f = \varphi(\mathbf{x}_i, t_i; t_f)$ be the propagated final state using the PBRFBP dynamic model. Let $\mathbf{x}(t) = \varphi(\mathbf{x}_i, t_i; t)$ be a solution of the equations of motion, integrated from $(\mathbf{x}_i, t_i)$ to $t \geq t_i$. The optimization problem consists of finding $\{\mathbf{x}_i, t_i, t_f\}$, with $t_f > t_i$, such that $\psi_i(\mathbf{x}_i) = 0$, $\psi_f(\mathbf{x}_f) = 0$, where $\mathbf{x}_f = \varphi(\mathbf{x}_i, t_i; t_f)$, and minimizing the cost function:

$$\Delta v(\mathbf{x}_i, t_i, t_f) = \Delta v_i(\mathbf{x}_i) + \Delta v_f(\varphi(\mathbf{x}_i, t_i; t_f)). \quad (8)$$

where the equality constraints ψ to be satisfied are:

$$\psi_i(\mathbf{x}_i) = \begin{cases} (x_i + \mu)^2 + y_i^2 - r_i^2 = 0, \\ (x_i + \mu)(\dot{x}_i - y_i) + y_i(\dot{y}_i + x_i + \mu) = 0, \end{cases} \quad (9a)$$

$$\psi_f(\mathbf{x}_f) = \begin{cases} (x_f + \mu - 1)^2 + y_f^2 - r_f^2 = 0, \\ (x_f + \mu - 1)(\dot{x}_f - y_f) + y_f(\dot{y}_f + x_f + \mu - 1) = 0. \end{cases} \quad (9b)$$

and the costs of initial and final manoeuvres are:

$$\Delta v_i = \sqrt{(\dot{x}_i - y_i)^2 + (\dot{y}_i + x_i + \mu)^2} - \sqrt{\frac{1 - \mu}{r_i}}, \quad (10a)$$

$$\Delta v_f = \sqrt{(\dot{x}_f - y_f)^2 + (\dot{y}_f + x_f + \mu - 1)^2} - \sqrt{\frac{\mu}{r_f}} \quad (10b)$$

The problem is solved as a non-linear programming (NLP) problem using the MATLAB function `fmincon`, which minimizes the total Δv (cost function), while satisfying the non-linear equality constraints ψ defined in Eq. (9). In addition, the linear inequality constraint, $t_f > t_i$, is included to ensure physical solutions. The initial guess consists in a vector that includes the initial state $\mathbf{x}_0$ derived from the solution in Section 2.1, augmented with the initial and final time $t_i, t_f = t_i + \delta$ in dimensionless units.

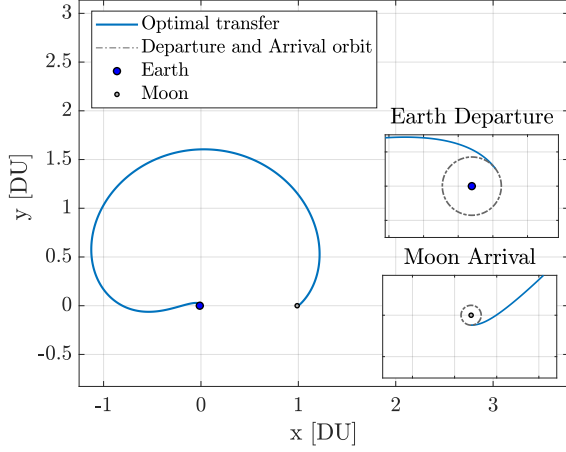
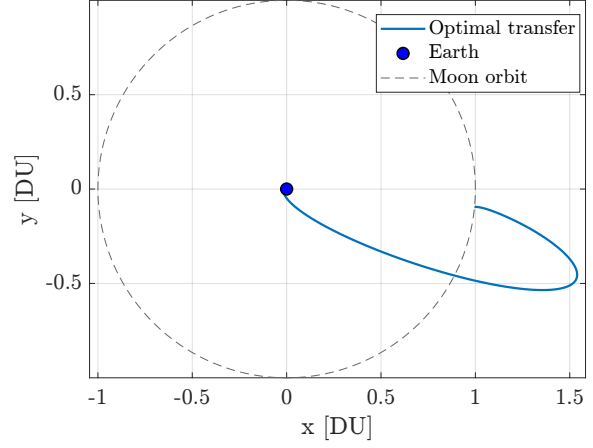
To ensure precise results, the `active-set` algorithm was utilized, with both the `ConstraintTolerance` and `OptimalityTolerance` parameters set to 10^{-10} .

The method is implemented in two cases: the first (a), without providing any derivative information to the solver, which computes itself the derivatives by using the forward finite differences. Then the procedure is repeated (b) providing the analytic derivatives of the objective function and constraints with respect to the NLP variables. The derivations are implemented by integrating the state transition matrix with a variational approach. For the computation of the analytic derivatives, refer to Appendix A.

Results The optimal variables for the two cases are reported in Table 6. As expected, providing the derivatives to the solver improves the convergence of `fmincon`, reducing the number of total iterations and function evaluations per iteration. Moreover, the achieved solution optimizes the objective function effectively since the total ΔV decreases from 4.0071 to 4.0068 [VU].

The initial state obtained is then propagated from t_0 to t_f . This time, as shown in Fig. 6 and Fig. 7, the propagation reaches the target orbit around the Moon with good precision.

Gradients	$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]	t_i [TU]	t_f [TU]
False	0.001578	0.010072	-6.327583	8.624902	2.085787	6.096448
True	0.001620	0.010013	-6.290741	8.651415	2.198151	6.203204

Table 6: Simple shooting solutions in the Earth-Moon rotating frame.**Figure 6:** Simple shooting transfer solution (@EMB Earth-Moon rotating frame).**Figure 7:** Simple shooting transfer solution (@Earth ECI).

2.3 Two-Impulse Transfer Optimization with Multiple Shooting

Methodology In this section, the Earth-Moon transfer is optimized using a multiple shooting technique. A time grid of $N = 4$ equally spaced points between t_i and t_f is defined. The state presented in Section 2.2 is augmented by introducing the $\mathbf{x}_j$ state vector of the satellite at the t_j epoch, where $t_j \in [t_1, t_2, \dots, t_N]$. Hence, the augmented state becomes $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, t_i, t_f\}$. The initial guesses $\mathbf{x}_j$ are achieved by propagating the guess used for the simple shooting from t_0 to t_j .

The new NLP problem is stated as:

$$\min_{\mathbf{y}} f(\mathbf{y}) \quad \text{subject to} \quad \mathbf{c}(\mathbf{y}) = 0, \quad \mathbf{g}_j(\mathbf{y}) < 0, \quad j = 1, \dots, 2N + 1. \quad (11)$$

Here, $\mathbf{c}(\mathbf{y})$ is defined as:

$$\mathbf{c}(\mathbf{y}) = \{\psi_1, \zeta_1, \dots, \zeta_{N-1}, \psi_N\}. \quad (12)$$

The term ψ_1 represents the constraints at the initial state on matching conditions of the departure orbit. Similarly, ψ_N represents the constraints at the final state. The vector ζ_j represents the defects between the propagated state starting from $\mathbf{x}_j$ and the state at the following node $\mathbf{x}_{j+1}$, and is computed as:

$$\zeta_j = \varphi(\mathbf{x}_j, t_j; t_{j+1}) - \mathbf{x}_{j+1}, \quad j = 1, \dots, N - 1. \quad (13)$$

The nonlinear inequality constraints $\mathbf{g}(\mathbf{y})$ are defined as:

$$\mathbf{g}(\mathbf{y}) = \{\eta_1, \dots, \eta_N\}, \quad (14)$$

where η_j , defined in Eq. (15), ensures that the trajectory avoids collisions with celestial bodies. Finally, the linear inequality constraint, $t_f > t_i$, is included, to ensure physical solutions.

$$\eta_j = \begin{cases} \left(\frac{R_e}{DU}\right)^2 - (x_j + \mu)^2 + y_j^2 < 0, \\ \left(\frac{R_m}{DU}\right)^2 - (x_j + \mu - 1)^2 + y_j^2 < 0, \end{cases} \quad j = 1, \dots, N. \quad (15)$$

The problem is solved again using `fmincon`, with the `ConstraintTolerance` set to 10^{-7} to ensure a balance between accuracy and computational cost. To enhance convergence, the analytic derivatives of both the objective function and the constraints with respect to the NLP variables are supplied to the solver. The complete derivation is provided in Appendix B.

Results With a total adimensionalized cost of $\Delta v = 3.9973$ [VU], the optimization using the multiple shooting technique demonstrates further improvement over the simple shooting method. This result was expected, as it confirms that the multiple shooting approach provides superior accuracy in handling long transfer times and nonlinear dynamics.

Fig. 8 highlights the computed transfer trajectory, showing the prescribed nodes along the path. The optimal initial state is presented in Table 7.

This solution could potentially be improved by reducing the constraint tolerance or increasing the number of nodes, though this would come at the cost of additional computational effort.

$r_{x,0}$ [DU]	$r_{y,0}$ [DU]	$v_{x,0}$ [VU]	$v_{y,0}$ [VU]	t_i [TU]	t_f [TU]
0.004072	0.005172	-3.248370	10.189172	2.462282	6.127306

Table 7: Multiple shooting solution in the Earth-Moon rotating frame.

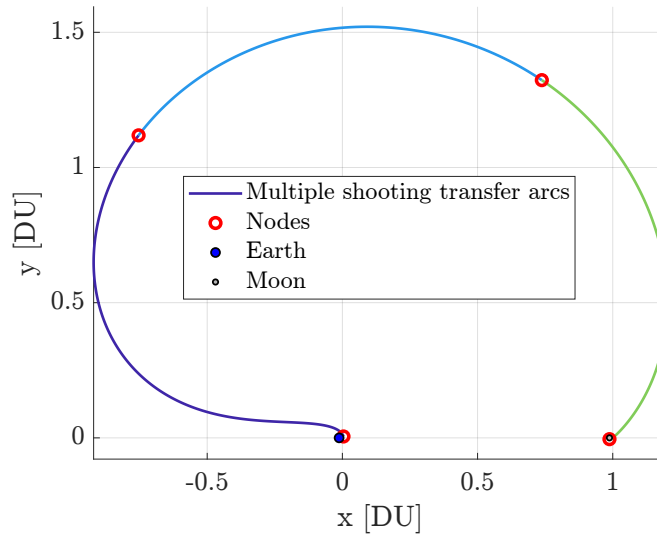


Figure 8: Multiple shooting transfer solution (@EMB Earth-Moon rotating frame)

2.4 Epoch Determination and N-Body Propagation

To determine the epoch where the Moon-EMB-Sun angle equals θ_i , a zero-finding approach is implemented. The initial value for θ_i is calculated as:

$$\theta_i = \omega_s \cdot t_i \quad (16)$$

where t_i is the adimensionalized departure time computed in Section 2.2(b), and ω_s is the scaled angular velocity of the Sun. The Earth-Moon orbital period T_{EM} , is computed from the Earth-Moon angular velocity ω_{EM} .

The starting epoch is set to 2024 Sep 28 00:00:00.000 TDB, which is converted into ephemeris time using the provided kernel. The search epoch interval is defined as the starting epoch plus one Earth-Moon orbital period to guarantee the unicity of the solution. To retrieve the target epoch, a zero-finding problem is implemented. The objective is to reduce to zero the difference between the reference angle θ_i and the angle computed from the kernel at each time belonging to the search epoch vector.

The problem is solved using the MATLAB function `fzero`. The cost function outline is summarized in Algorithm 1.

Algorithm 1 Cost function: Theta finder algorithm

Require: t : Time epoch, θ_i : Initial angular displacement

Ensure: θ_f : Final angular displacement

Retrieve Moon and Sun position in the J2000 frame with SPICE:

$\mathbf{r}_{\text{Moon}}, \mathbf{r}_{\text{Sun}} \leftarrow \text{cspice_spkpos}$

Compute the angular position of Moon and Sun:

$\theta_{\text{Moon}}, \theta_{\text{Sun}} \leftarrow \text{atan2}(r_y, r_x)$

Calculate the angular difference:

$\theta \leftarrow \theta_{\text{Sun}} - \theta_{\text{Moon}}$

Compute the final angular displacement:

$\theta_f \leftarrow \theta - \theta_i$

return θ_f

The time of flight (ToF) between t_i and t_f is computed based on the adimensionalized transfer solution and the arrival epoch is retrieved from kernel.

$$\text{ToF} = (t_f - t_i) \cdot \text{TU} \cdot 86400, \quad (17)$$

The initial state in the Earth-Moon rotating frame is transformed into the Earth-Centered Inertial (ECI) frame using a coordinate transformation routine.

The transfer is now generalized to a 3D N-body propagation following newtonian motion only. The equation of motion of the N-body propagator can be expressed as:

$$\begin{bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{a} \end{bmatrix} \quad (18)$$

The vector $\mathbf{a}$ is the total gravitational acceleration, given by $\mathbf{a} = \mathbf{a}_0 + \sum_{i \neq 0} \mathbf{a}_i$. $\mathbf{a}_0$ accounts for the main attractor acceleration while $\mathbf{a}_i$ is the contribution coming from the i -th perturbing body.

$$\mathbf{a}_0 = -\frac{\mu_{\text{main}} \mathbf{r}}{r^3} \quad \mathbf{a}_i = -\frac{\mu_i}{|\mathbf{r} - \boldsymbol{\rho}_i|^3} (\mathbf{r} + \boldsymbol{\rho}_i f(q)) \quad (19)$$

Where $\boldsymbol{\rho}_i$ is the relative position of the perturbing body with respect to the center body, and the perturbation function $f(q)$ is defined as:

$$f(q) = \frac{q(3 + 3q + q^2)}{1 + (1 + q)^{3/2}}, \quad \text{with} \quad q = \frac{\mathbf{r} \cdot (\mathbf{r} - 2\boldsymbol{\rho}_i)}{|\boldsymbol{\rho}_i|^2} \quad (20)$$

To perform propagation, the initial state is enlarged by adding the out-of-plane components for both position and velocity $z_i = 0$, $v_{z,i} = 0$. The results are compared graphically with the shooting solutions obtained in the previous sections.

Results The starting and final epochs obtained from the simulation are summarized in Table 8, along with the initial state vector in the Earth-centered inertial (ECI) frame. The results for the initial state vector are expressed in physical units.

Fig. 9 compares the propagation results from the planar bicircular four-body problem (PCBRFBP) with the trajectory obtained with the 3D N -body problem. As observed, the PCBRFBP fails to reach the target orbit due to the differences in the dynamical models employed. Furthermore, although the in-plane positions (x - y) are still quite similar to those obtained from the PCBRFBP, the addition of the 3D nature to the N -body propagator causes noticeable variations along the z -axis. This highlights the limitations of the planar approximation.

Symbol	Calendar epoch (UTC)		
t_i	2024-10-12-T-11:40:34.1658		
t_f	2024-10-29-T-21:05:20.3260		
$r_{x,0}$ [km]	$r_{y,0}$ [km]	$r_{z,0}$ [km]	
-6223.6284	2026.1447	0.0000	
$v_{x,0}$ [km/s]	$v_{y,0}$ [km/s]	$v_{z,0}$ [km/s]	
-3.3980	-10.4375	0.0000	

Table 8: Initial epoch, final epoch, and initial state in Earth-centered inertial frame.

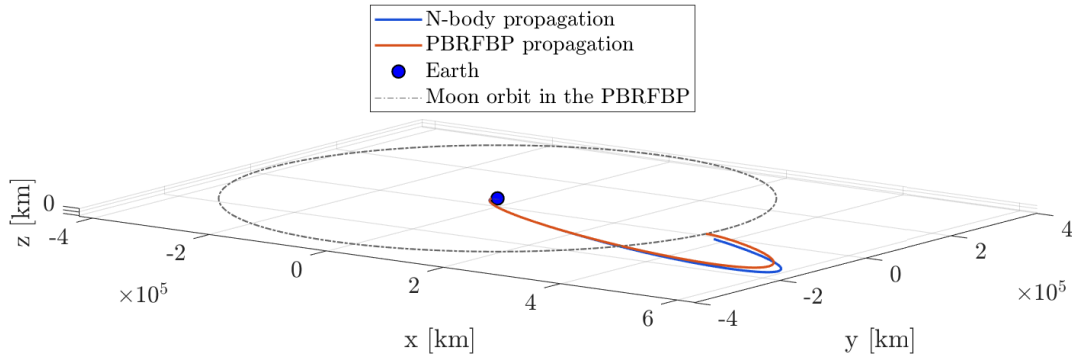


Figure 9: Comparison between the 3D N -body propagator and the PBRFBP propagation. (@Earth ECI)

3 Continuous guidance

Exercise 3

A low-thrust option is being considered to perform an orbit raising maneuver using a low-thrust propulsion system in Earth orbit. The spacecraft is released on a circular orbit on the equatorial plane at an altitude of 800 km and has to reach an orbit inclined by 0.75 deg on the equatorial plane at 1000 km. This orbital regime is characterized by a large population of resident space objects and debris, whose spatial density q can be expressed as:

$$q(\rho) = \frac{k_1}{k_2 + \left(\frac{\rho - \rho_0}{DU}\right)^2}$$

where ρ is the distance from the Earth center. The objective is to design an optimal orbit raising that minimizes the risk of impact, that is to minimize the following objective function

$$F(t) = \int_{t_0}^{t_f} q(\rho(t)) dt.$$

The parameters and reference Distance Unit to be considered are provided in Table 9.

Symbol	Value	Units	Meaning
h_i	800	km	Altitude of departure orbit
h_f	1000	km	Altitude of arrival orbit
Δi	0.75	deg	Inclination change
R_e	6378.1366	km	Earth radius
μ	398600.435	km ³ /s ²	Earth gravitational parameter
ρ_0	$750 + R_e$	km	Reference radius for debris flux
k_1	1×10^{-5}	DU ⁻¹	Debris spatial density constant 1
k_2	1×10^{-4}	DU ²	Debris spatial density constant 2
m_0	1000	kg	Initial mass
$T_{\max}$	3.000	N	Maximum thrust
I_{sp}	3120	s	Specific impulse
DU	7178.1366	km	Distance Unit
MU	m_0	kg	Mass Unit

Table 9: Problem parameters and constants. The units of time TU and velocity VU can be computed imposing that the scaled gravitational parameter $\bar{\mu} = 1$.

- 1) Plot the debris spatial density $q(\rho) \in [h_i - 100; h_f + 100]$ km and compute the initial state and target orbital state, knowing that: i) the initial and final state are located on the x -axis of the equatorial J2000 reference frame; ii) the rotation of the angle Δi is performed around the x -axis of the equatorial J2000 reference frame (RAAN = 0).
- 2) Adimensionalize the problem using as reference length $DU = \rho_i = h_i + R_e$ and reference mass $MU = m_0$, imposing that $\mu = 1$. Report all the adimensionalized parameters.
- 3) Using the PMP, write down the spacecraft equations of motion, the costate dynamics, and the zero-finding problem for the unknowns $\{\lambda_0, t_f\}$ with the appropriate transversality condition. **Hint:** the spacecraft has to reach the target state computed in point 1).
- 4) Solve the problem considering the data provided in Table 9. To obtain an initial guess for the costate, generate random numbers such that $\lambda_{0,i} \in [-250; +250]$ while $t_f \approx 20\pi$.

Report the obtained solution in terms of $\{\lambda_0, t_f\}$ and the error with respect to the target. Assess your results exploiting the properties of the Hamiltonian in problems that are not time-dependent and time-optimal solution. Plot the evolution of the components of the primer vector α in a NTW reference frame[†].

- 5) Solve the problem for a lower thrust level $T_{\max} = 2.860$ N. Compare the new solution with the one obtained in the previous point. **Hint:** exploit numerical continuation.

(11 points)

[†]The T-axis is aligned with the velocity, the N-axis is aligned with the angular momentum, while the W-axis is pointing inwards, i.e., towards the Earth.

3 Continuous Guidance

3.1 Debris Spatial density and Boundary Conditions

Methodology The debris spatial density $q(\rho)$ was computed using the provided equation across the prescribed altitude range. The initial state vector was constructed through simple orbital mechanics consideration.

$$\mathbf{r}_i = [r_i, 0, 0]^T \quad \mathbf{v}_i = [0, \sqrt{\mu/r_i}, 0]^T \quad (21)$$

where $r_i = h_i + R_e$ represents the initial orbital radius, μ is Earth's gravitational parameter and R_e is the Earth's radius.

Similarly, the final state was computed considering the rotation around the x -axis to change inclination of $\Delta i = 0.75^\circ$. The target position and velocity components were expressed as:

$$\mathbf{r}_f = [r_f, 0, 0]^T \quad \mathbf{v}_f = [0, \sqrt{\mu/r_f} \cdot \cos(\Delta i), \sqrt{\mu/r_f} \cdot \sin(\Delta i)]^T \quad (22)$$

where $r_f = h_f + R_e$ represents the final orbital radius.

Results The debris spatial density $q(\rho)$ is plotted against altitude in Fig. 10, exhibiting a peak near to the reference radius ρ_0 and an exponential decline for higher altitudes. Table 10 provides a summary of the initial and target state vectors, expressed in the equatorial J2000 reference frame.

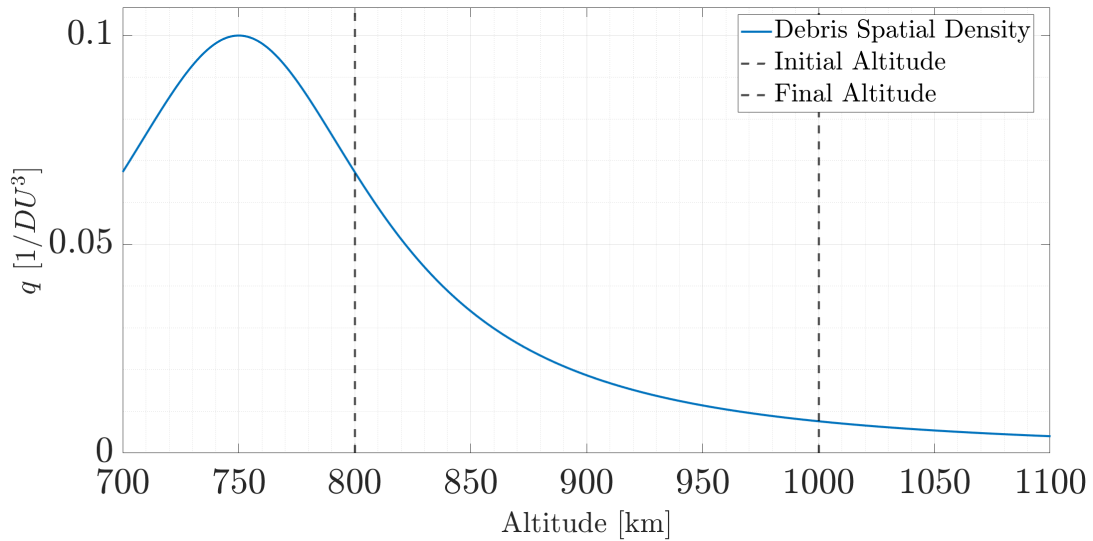


Figure 10: Debris spatial density as a function of altitude.

$r_{x,i}$ [km]	$r_{y,i}$ [km]	$r_{z,i}$ [km]	$v_{x,i}$ [km/s]	$v_{y,i}$ [km/s]	$v_{z,i}$ [km/s]
7178.136600	0.000000	0.000000	0.00000000	7.45183148	0.00000000
$r_{x,f}$ [km]	$r_{y,f}$ [km]	$r_{z,f}$ [km]	$v_{x,f}$ [km/s]	$v_{y,f}$ [km/s]	$v_{z,f}$ [km/s]
7378.136600	0.000000	0.000000	0.00000000	7.34950906	0.09621034

Table 10: Initial and target state in Earth-centered equatorial J2000 inertial frame.

Point 2: Adimensionalization

Methodology The adimensionalization is carried out starting from the given reference units DU and MU , and the values of TU and VU are computed by imposing $\bar{\mu} = 1$:

$$TU = \sqrt{\frac{DU^3}{\mu}}, \quad VU = \frac{DU}{TU} \quad (23)$$

Then, all the problem variables are adimensionalized through straightforward computations.

Results From the values of $TU = 963.271462 [s]$ and $VU = 7.451831 [km/s]$, the adimensionalized parameters are reported in Table 11, while the adimensional state is presented in Table 12.

$h_i [DU]$	$h_f [DU]$	$\Delta i [rad]$	$R_E [DU]$	$\mu [DU^3/TU^2]$	
0.111450	0.139312	± 0.013090	0.888550	1.000000	

$m_0 [MU]$	$T_{\max} [MU \ DU/TU^2]$	$I_{sp} [TU]$	$g_0 [DU/TU^2]$	$\rho_0 [DU]$	
1.000000	0.000388	3.238962	1.268103	0.993034	

Table 11: Adimensionalized parameters in the problem.

$r_{x,i} [DU]$	$r_{y,i} [DU]$	$r_{z,i} [DU]$	$v_{x,i} [VU]$	$v_{y,i} [VU]$	$v_{z,i} [VU]$
1.000000	0.000000	0.000000	0.00000000	1.00000000	0.00000000

$r_{x,f} [DU]$	$r_{y,f} [DU]$	$r_{z,f} [DU]$	$v_{x,f} [VU]$	$v_{y,f} [VU]$	$v_{z,f} [VU]$
1.027862	0.000000	0.000000	0.00000000	0.986269	0.012911

Table 12: Initial and target states in Earth-centered equatorial J2000 inertial frame (adimensionalized).

3.2 PMP Formulation

Methodology In this section, the problem is formulated using the Ponderagin Maximum Principle. The objective is to minimize the risk of collision with resident space debris during the orbit raising maneuver. The problem is stated as follows:

$$\min_{\mathbf{u} \in \mathcal{U}} \int_{t_0}^{t_f} \ell(\mathbf{x}, \mathbf{u}, t) dt \quad \text{s.t.} \quad \begin{cases} \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, t), \\ \mathbf{x}(t_0) = \mathbf{x}_0, \\ \mathbf{x}(t_f) = \mathbf{x}_f. \end{cases} \quad (24)$$

where $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, t)$ is the low-thrust dynamics and $\mathbf{x}(t_0) = \mathbf{x}_0$, $\mathbf{x}(t_f) = \mathbf{x}_f$ are the boundary conditions, stated as:

$$\begin{cases} \dot{\mathbf{r}} = \mathbf{v}, \\ \dot{\mathbf{v}} = -\frac{\mu}{r^3} \mathbf{r} + u \frac{T_{\max}}{m} \boldsymbol{\alpha}, \\ \dot{m} = -u \frac{T_{\max}}{I_{sp} g_0} \end{cases} \quad \begin{cases} \mathbf{r}(t_0) = \mathbf{r}_0, \\ \mathbf{v}(t_0) = \mathbf{v}_0, \\ \mathbf{r}(t_f) = \mathbf{r}_f, \\ \mathbf{v}(t_f) = \mathbf{v}_f, \\ m(t_0) = m_0, \\ m(t_f) = m_f, \\ \lambda_m(t_f) = 0 \end{cases} \quad (25)$$

with $\mathcal{U}$ be the set of admissible control conditions, $\hat{\alpha}$ the prime vector and accounts for the thrust direction, and u the engine throttle level.

$$\mathcal{U} = \{(u, \hat{\alpha}), \quad \text{s.t.} \quad u \in [0, 1], \quad \hat{\alpha} = 1\} \quad (26)$$

The goal of the PMP is to minimize the Hamiltonian, H , at all times, within the admissible control values. The Hamiltonian is formulated as:

$$H(\mathbf{x}, \mathbf{u}, \boldsymbol{\lambda}, t) = \ell(\mathbf{x}, \mathbf{u}, t) + \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{x}, \mathbf{u}, t). \quad (27)$$

The objective function is defined as $\ell(\mathbf{x}, \mathbf{u}, t) = q(\rho)$, while $\boldsymbol{\lambda}$ represents the lagrange multipliers (co-state). Hence Eq. (27) could be written as:

$$H = \ell + \boldsymbol{\lambda}^T \mathbf{f} = q(\rho) + \begin{pmatrix} \lambda_r \\ \lambda_v \\ \lambda_m \end{pmatrix}^T \begin{pmatrix} \mathbf{v} \\ -\frac{\mu}{r^3} \mathbf{r} + \frac{T_{\max}}{m} \hat{\alpha} \\ -\frac{T_{\max}}{I_{\text{sp}} g_0} \end{pmatrix} \quad (28)$$

The problem is solved as a time optimal problem, hence the thrust throttle level u is always set to 1. The prime vector $\hat{\alpha}^*$, makes sure that the thrust direction is in line with the velocity co-state's negative gradient.

$$\hat{\alpha}^* = -\frac{\boldsymbol{\lambda}_v}{\|\boldsymbol{\lambda}_v\|} \quad (29)$$

From the Hamiltonian the Euler-Lagrange Equations could be derived as follows:

$$\begin{cases} \dot{\mathbf{x}} = \frac{\partial H}{\partial \boldsymbol{\lambda}} \\ \dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{x}} \end{cases} \rightarrow \begin{cases} \dot{\mathbf{r}} = \mathbf{v} \\ \dot{\mathbf{v}} = -\frac{\mu}{r^3} \mathbf{r} - \frac{T_{\max}}{m} \frac{\boldsymbol{\lambda}_v}{\|\boldsymbol{\lambda}_v\|} \\ \dot{m} = -\frac{T_{\max}}{I_{\text{sp}} g_0} \\ \dot{\lambda}_r = -\frac{\partial q(\rho)}{\partial \mathbf{r}} - 3 \frac{\mu}{r^5} (\mathbf{r} \cdot \boldsymbol{\lambda}_v) \mathbf{r} + \frac{\mu}{r^3} \boldsymbol{\lambda}_v \\ \dot{\lambda}_v = -\lambda_r \\ \dot{\lambda}_m = -\lambda_v \frac{T_{\max}}{m^2} \end{cases} \quad (30)$$

Since the arrival t_f is unknown, the system is closed by providing the proper transversality condition. In this problem the target is fixed hence $\dot{\psi}(t_f) = 0$.

$$H(t_f) - \boldsymbol{\lambda}(t_f) \cdot \dot{\psi}(t_f) = 0 \quad \rightarrow \quad H(t_f) = 0 \quad (31)$$

The objective is to find λ_0 and t_f such that the propagated final state and costate result in an optimal solution that meets both the transversality and boundary conditions.

3.3 Numerical Solution to the PMP

Methodology The initial guess for the costate vector, $\boldsymbol{\lambda}_0$, was generated at random, ensuring $\lambda_{0,i} \in [-250, 250]$. Negative values are excluded for the last element of the co-state λ_m . The initial estimate for the final time, t_f , was set to $t_f \approx 20\pi$, by slightly perturbing this reference value.

The zero-finding problem was solved using the MATLAB function `fsolve` with the `levenberg-marquardt` algorithm. This algorithm was chosen because of its ability to determine a solution given an uncertain initial guess. To achieve faster convergence, the objective

Algorithm 2 Cost Function for Initial Costate and Final Time Estimation**Require:** $[\lambda_0, t_f]$: Initial guess for costate vector and final time, $[\mathbf{x}_{\text{target}}; 0; 0]$: Target final conditions. $\mathbf{x}_0$: Initial state vector,**Ensure:** $\mathbf{e}_{\text{vec}}$: Error vector.Initialize state vector and costate $[\mathbf{x}_0; \lambda_0]$ Initialize state transition matrix Φ_0 as an identity matrix $\mathbf{I}_{14 \times 14}$ Construct complete initial state by appending Φ_0 to $[\mathbf{x}_0; \lambda_0]$ **Propagate System and State Transition Matrix:**Propagate the state from t_0 to t_f by integrating the Euler-Lagrange equations and STMExtract final state vector $[\mathbf{x}_f; \lambda_f]$ from the propagation resultsExtract final state transition matrix Φ_f from the propagation results**Compute Hamiltonian:**Evaluate the Hamiltonian H at the final state $[\mathbf{x}_f; \lambda_f]$ **Compute the Derivatives:**Use the final state transition matrix Φ_f and final state to compute the gradient ∇f **Compute Error Vector:** $\mathbf{e}_{\text{vec}} := [\Delta \mathbf{x}_f; \Delta \lambda_{m,f}; \Delta H(t_f)] = [\mathbf{x}_f; \lambda_{m,f}; H(t_f)] - [\mathbf{x}_{\text{target}}; 0; 0]$ **return** $\mathbf{e}_{\text{vec}}, \nabla f$

function's derivatives was given to the solver. Appendix C describes the process for obtaining these derivatives. Algorithm 2 provides the cost function's pseudocode in detail.

The primer vector $\hat{\alpha}^*$ is computed as stated in Eq. (29) and its components were rotated into the NTW reference frame through:

$$\hat{\alpha}_{\text{NTW}} = \mathbf{R} \cdot \hat{\alpha}_{\text{ECI}}, \quad (32)$$

with:

$$\mathbf{R} = [\hat{\mathbf{N}}, \hat{\mathbf{T}}, \hat{\mathbf{W}}], \quad \hat{\mathbf{N}} = \frac{\mathbf{r}_{\text{ECI}} \times \mathbf{v}_{\text{ECI}}}{\|\mathbf{r}_{\text{ECI}} \times \mathbf{v}_{\text{ECI}}\|}, \quad \hat{\mathbf{T}} = \frac{\mathbf{v}_{\text{ECI}}}{\|\mathbf{v}_{\text{ECI}}\|}, \quad \hat{\mathbf{W}} = \hat{\mathbf{N}} \times \hat{\mathbf{T}} \quad (33)$$

The achieved state and co-state were propagated using `ode113` in MATLAB, integrating the Euler-Lagrange equations. The absolute errors between the propagating and target states are then estimated for location and velocity, respectively.

Finally, the Hamiltonian was evaluated at each point along the transfer trajectory to assess the validity of the solution.

Results The optimized costate vector λ_0 are reported in Table 13 with the final time t_f and the final mass m_f . As prescribed, the final time is close the first guess value, providing a transfer which lasts around 10 orbital period to reach the target state.

In Fig. 12 is showcased the transfer trajectory obtained from the propagation. A clearer representation is provided in Fig. 11, where the reference axes are intentionally scaled differently to highlight the plane change manoeuvre and the transfer orbit.

The error between the final state and the target state is reported in Table 14. The errors on both velocity and position are remarkably small, indicating that the numerical solution is highly accurate. This implies that the implemented method effectively catches the system dynamics.

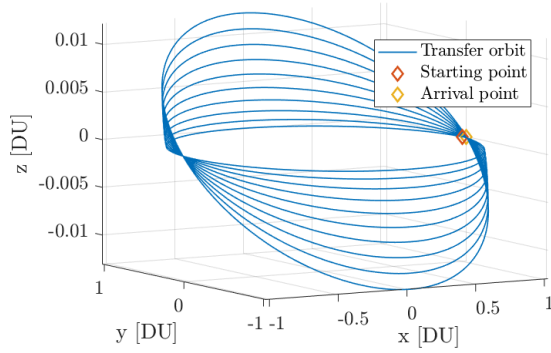
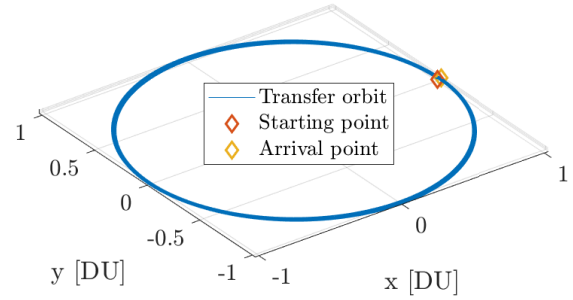
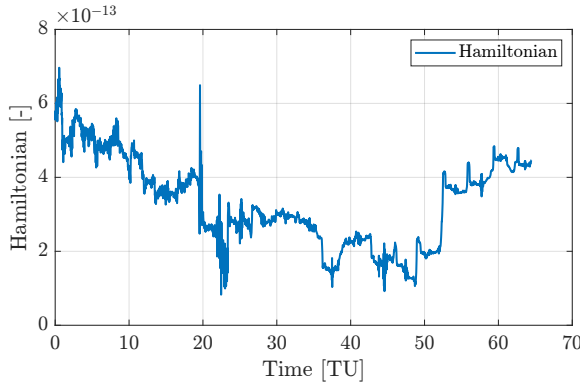
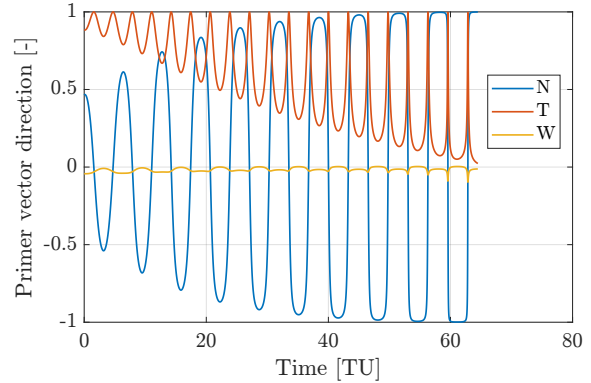
The Hamiltonian over time, shown in Fig. 13, remained consistently in the range of 10^{-12} to 10^{-13} , assessing the validity of the solution found.

Fig. 14 resumes the evolution of the primer vector in the NTW frame. The T-component is always positive and oscillates with a growing trend, indicating a thrust alignment with the positive velocity direction, as required for orbit raising. Furthermore, the N-component oscillates between positive and negative values at a frequency comparable to the number of orbital periods during propagation. This reflects the inclination change provided by the maneuver.

t_f [mins]		m_f [kg]				
1035.1974		993.9120				
λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
-214.9812	-10.3659	0.8856	-10.3929	-214.6105	-112.9454	2.5964

Table 13: Optimal orbit raising transfer solution ($T_{\max} = 3.000$ N).

Error	Value	Units
$\ \mathbf{r}(t_f) - \mathbf{r}_f\ $	$3.4252 \cdot 10^{-8}$	km
$\ \mathbf{v}(t_f) - \mathbf{v}_f\ $	$1.6242 \cdot 10^{-7}$	m/s

Table 14: Final state error with respect to target position and velocity ($T_{\max} = 3.000$ N).**Figure 11:** Transfer trajectory with scaled axis, $T_{\max} = 3.000$ N (@Earth ECI).**Figure 12:** Transfer trajectory with equal axis, $T_{\max} = 3.000$ N (@Earth ECI).**Figure 13:** Hamiltonian over time during the propagation.**Figure 14:** Evolution of the primer vector components in NTW frame $T = 3$ N.

3.4 Lower Thrust Solution with Numerical Continuation

Methodology . The reduced thrust value of $T_{\max} = 2.86$ N is adimensionalized. To exploit numerical continuation, a vector of decreasing equally spaced thrust values is defined, ranging from $T_{\max} = 3.00$ N to $T_{\max} = 2.86$ N. The algorithm iteratively corrects the initial state at each step, starting from the solution found in Section 3.3. To solve the zero-finding problem, the MATLAB `fsolve` function is used with the `trust-region-dogleg` algorithm, which is particularly efficient when the initial guess is close to the solution.

The achieved final state and costate system of equations was propagated using `ode113` in MATLAB by integrating the EL equation. To assess the validity of the solution, the Hamiltonian

is evaluated for every point of the transfer.

Results The new initial solution obtained for $T_{\max, \text{new}} = 2.86 \text{ N}$ is reported in Table 15 together with the final time and the final mass. The error in both position and velocity on the target orbit is summarized in Table 16.

The solution obtained with the reduced thrust level improves the matching with the final conditions, but this result appears to be case-dependent. By shooting the initial condition multiple times, the errors obtained with the two thrust levels oscillate between 10^{-7} and 10^{-10} lacking a consistent relationship between the two methods. Conversely, lower thrust levels always result in lower fuel consumption, as the final mass is greater than in the previous case for each trial. This behavior may be explained by the differences in the evolution of the primer vector with $T = 2.86$. As presented in Fig. 14, the N and T components exhibit a more regular trend, while the W component stays closer to zero compared to the first case.

t_f [mins]		m_f [kg]				
1030.9760		994.2198				
λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
-593.1529	-11.5904	2.2973	-11.9293	-592.7890	-920.3698	17.3814

Table 15: Optimal orbit raising transfer solution ($T_{\max} = 2.860 \text{ N}$).

Error	Value	Units
$\ \mathbf{r}(t_f) - \mathbf{r}_f\ $	$8.8959 \cdot 10^{-10}$	km
$\ \mathbf{v}(t_f) - \mathbf{v}_f\ $	$7.5275 \cdot 10^{-10}$	m/s

Table 16: Final state error with respect to target position and velocity ($T_{\max} = 2.860 \text{ N}$).

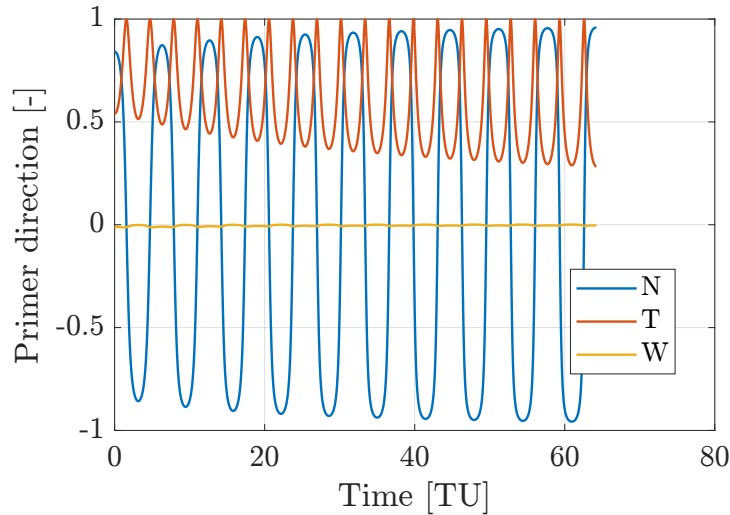


Figure 15: Evolution of the primer vector components in NTW frame $T = 2.86$.

Appendix

A Analytical Derivatives Simple Shooting

The derivative of the objective function J with respect to the NLP variables: $\nabla J(\mathbf{x}) = \left[\frac{\partial J}{\partial \mathbf{x}_0}, \frac{\partial J}{\partial t_i}, \frac{\partial J}{\partial t_f} \right]^T$, and of the constraints, $\nabla \mathbf{c}(\mathbf{x})$, are evaluated as follows:

The spatial derivatives of the objective function $J(\mathbf{x})$ are expressed as:

$$\frac{\partial J}{\partial \mathbf{x}_0} = \frac{\partial J}{\partial \mathbf{x}_i} + \frac{\partial J}{\partial \mathbf{x}_f} \Phi(t_0, t_f)$$

with:

$$\begin{aligned} \frac{\partial J}{\partial \mathbf{x}_i} &= \frac{1}{\sqrt{(\dot{x}_i - y_i)^2 + (\dot{y}_i + x_i + \mu)^2}} [\dot{y}_i + x_i + \mu, y_i - \dot{x}_i, \dot{x}_i - y_i, \dot{y}_i + x_i + \mu], \\ \frac{\partial J}{\partial \mathbf{x}_f} &= \frac{1}{\sqrt{(\dot{x}_f - y_f)^2 + (\dot{y}_f + x_f + \mu - 1)^2}} [\dot{y}_f + x_f + \mu - 1, y_f - \dot{x}_f, \dot{x}_f - y_f, \dot{y}_f + x_f + \mu - 1]. \end{aligned}$$

While the time derivatives are calculated as

$$\frac{dJ}{dt_i} = -\frac{\partial J}{\partial \mathbf{x}_f} \Phi(t_0, t_f) \mathbf{f}(\mathbf{x}_0, t_i) \quad ; \quad \frac{dJ}{dt_f} = \frac{\partial J}{\partial \mathbf{x}_f} \mathbf{f}(\mathbf{x}_f, t_f)$$

As for the constraints:

$$\begin{aligned} \frac{\partial \psi_i}{\partial \mathbf{x}_i} &= \begin{bmatrix} 2(x_i + \mu) & 2y_i & 0 & 0 \\ \dot{x}_i & \dot{y}_i & (x_i + \mu) & y_i \end{bmatrix}, \\ \frac{\partial \psi_f}{\partial \mathbf{x}_f} &= \begin{bmatrix} 2(x_f + \mu - 1) & 2y_f & 0 & 0 \\ \dot{x}_f & \dot{y}_f & (x_f + \mu - 1) & y_f \end{bmatrix}. \end{aligned}$$

Obtaining:

$$\nabla \mathbf{c}(\mathbf{x}) = \begin{bmatrix} \frac{\partial \psi_i}{\partial x_i} & 0 & 0 \\ \frac{\partial \psi_f}{\partial x_f} \Phi_f & \frac{\partial \psi_f}{\partial x_f} (-\Phi(t_0, t_f) \mathbf{f}(t_i)) & \frac{\partial \psi_f}{\partial x_f} \cdot \mathbf{f}(t_f) \end{bmatrix}^T$$

where $\mathbf{f}$ is the right-hand side of the equations of motion.

To propagate the STM Φ with the variational approach such that:

$$\begin{cases} \dot{\Phi}(t_0, t) = \mathbf{A}(t) \Phi(t_0, t) \\ \Phi_0 = \mathbf{I}_{4 \times 4} \end{cases}$$

with th Jacobian matrix $\mathbf{A}(\mathbf{t})$ defined as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial \dot{x}} & \frac{\partial f_1}{\partial \dot{y}} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial \dot{x}} & \frac{\partial f_2}{\partial \dot{y}} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial \dot{x}} & \frac{\partial f_3}{\partial \dot{y}} \\ \frac{\partial f_4}{\partial x} & \frac{\partial f_4}{\partial y} & \frac{\partial f_4}{\partial \dot{x}} & \frac{\partial f_4}{\partial \dot{y}} \end{bmatrix}$$

B Analytical Derivatives for Multiple Shooting

Derivative of the Objective Function

The derivative of the objective function J with respect to the optimization variables $\mathbf{y}$ is expressed as:

$$\frac{\partial J}{\partial \mathbf{y}} = [\mathbf{P}_1 \quad \mathbf{O} \quad \mathbf{P}_N \quad \mathbf{O}],$$

where:

$$\mathbf{P}_1 := \frac{1}{\sqrt{(\dot{x}_i - y_i)^2 + (\dot{y}_i + x_i + \mu)^2}} [\dot{y}_i + x_i + \mu, y_i - \dot{x}_i, \dot{x}_i - y_i, \dot{y}_i + x_i + \mu],$$

$$\mathbf{P}_N := \frac{1}{\sqrt{(\dot{x}_f - y_f)^2 + (\dot{y}_f + x_f + \mu - 1)^2}} [\dot{y}_f + x_f + \mu - 1, y_f - \dot{x}_f, \dot{x}_f - y_f, \dot{y}_f + x_f + \mu - 1].$$

Jacobian of the Equality Constraints

The derivative of the equality constraints $\mathbf{c} = \{\zeta_j, \psi_1, \psi_N\}$ with respect to $\mathbf{y}$ is expressed as:

$$\frac{\partial \mathbf{c}}{\partial \mathbf{y}} = \begin{bmatrix} \Phi(t_1, t_2) & -\mathbf{I}_4 & \mathbf{O} & \mathbf{O} & \mathbf{Q}_1^1 & \mathbf{Q}_1^N \\ \mathbf{O} & \Phi(t_2, t_3) & -\mathbf{I}_4 & \mathbf{O} & \mathbf{Q}_2^1 & \mathbf{Q}_2^N \\ \mathbf{O} & \mathbf{O} & \Phi(t_3, t_4) & -\mathbf{I}_4 & \mathbf{Q}_3^1 & \mathbf{Q}_3^N \\ \mathbf{R}_1 & \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{R}_N & \mathbf{O} & \mathbf{O} \end{bmatrix},$$

where:

$$\mathbf{Q}_j^1 := -\frac{N-j}{N-1} \Phi(t_j, t_{j+1}) \mathbf{f}(\mathbf{x}_j, t_j) + \frac{N-j-1}{N-1} \mathbf{f}(\varphi(\mathbf{x}_j, t_j, t_j+1), t_{j+1}),$$

$$\mathbf{Q}_j^N := -\frac{j-1}{N-1} \Phi(t_j, t_{j+1}) \mathbf{f}(\mathbf{x}_j, t_j) + \frac{j}{N-1} \mathbf{f}(\varphi(\mathbf{x}_j, t_j, t_j+1), t_{j+1}),$$

$$\mathbf{R} := \frac{\partial \psi_i}{\partial \mathbf{x}_i} = \begin{bmatrix} 2(x_i + \mu) & 2y_i & 0 & 0 \\ \dot{x}_i & \dot{y}_i & (x_i + \mu) & y_i \end{bmatrix},$$

$$\mathbf{R}_N := \frac{\partial \psi_f}{\partial \mathbf{x}_f} = \begin{bmatrix} 2(x_f + \mu - 1) & 2y_f & 0 & 0 \\ \dot{x}_f & \dot{y}_f & (x_f + \mu - 1) & y_f \end{bmatrix}.$$

Inequality Constraints

The derivative of the inequality constraints $g(y) := \{\eta_j, \tau\} < 0$ with respect to the NLP variables y can be expressed as:

$$\frac{\partial \mathbf{g}}{\partial \mathbf{y}} = \begin{bmatrix} \mathbf{S}_1 & \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_2 & \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{S}_3 & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \ddots & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{S}_l \end{bmatrix} \quad \begin{aligned} \mathbf{S}_j &:= \frac{\partial \eta_j}{\partial \mathbf{x}_j} = \begin{bmatrix} -2(x_j + \mu) & -2y_j & 0 & 0 \\ -2(x_j + \mu - 1) & -2y_j & 0 & 0 \end{bmatrix}, \quad j = 1, \dots, N, \\ \mathbf{S}_t &:= \begin{bmatrix} \frac{\partial \tau}{\partial t_1} & \frac{\partial \tau}{\partial t_N} \end{bmatrix} = \begin{bmatrix} 1 & -1 \end{bmatrix}. \end{aligned}$$

C Analytical Derivatives for the PMP

Consider the objective function $\mathbf{F}$ and the optimization variables $\mathbf{y}$:

$$\mathbf{F} = \begin{pmatrix} \mathbf{r}(t_f) - \mathbf{r}_f \\ \mathbf{v}(t_f) - \mathbf{v}_f \\ \lambda_m(t_f) \\ H(t_f) \end{pmatrix}_{8 \times 1}, \quad \mathbf{y} = \begin{pmatrix} \lambda_0 \\ t_f \end{pmatrix}_{8 \times 1}$$

The derivative of the objective function $\mathbf{F}$ with respect to the optimization variables $\mathbf{y}$ is:

$$\frac{\partial \mathbf{F}}{\partial \mathbf{y}} = \begin{bmatrix} \frac{\partial \mathbf{r}(t_f)}{\partial \lambda_{\mathbf{r}0}} & \frac{\partial \mathbf{r}(t_f)}{\partial \lambda_{\mathbf{v}0}} & \frac{\partial \mathbf{r}(t_f)}{\partial \lambda_{\mathbf{m}0}} & \frac{\partial \mathbf{r}(t_f)}{\partial t_f} \\ \frac{\partial \mathbf{v}(t_f)}{\partial \lambda_{\mathbf{r}0}} & \frac{\partial \mathbf{v}(t_f)}{\partial \lambda_{\mathbf{v}0}} & \frac{\partial \mathbf{v}(t_f)}{\partial \lambda_{\mathbf{m}0}} & \frac{\partial \mathbf{v}(t_f)}{\partial t_f} \\ \frac{\partial m_f}{\partial \lambda_{\mathbf{r}0}} & \frac{\partial m_f}{\partial \lambda_{\mathbf{v}0}} & \frac{\partial m_f}{\partial \lambda_{\mathbf{m}0}} & \frac{\partial m_f}{\partial t_f} \\ \frac{\partial \lambda_r(t_f)}{\partial \lambda_{\mathbf{r}0}} & \frac{\partial \lambda_r(t_f)}{\partial \lambda_{\mathbf{v}0}} & \frac{\partial \lambda_r(t_f)}{\partial \lambda_{\mathbf{m}0}} & \frac{\partial \lambda_r(t_f)}{\partial t_f} \\ \frac{\partial \lambda_v(t_f)}{\partial \lambda_{\mathbf{r}0}} & \frac{\partial \lambda_v(t_f)}{\partial \lambda_{\mathbf{v}0}} & \frac{\partial \lambda_v(t_f)}{\partial \lambda_{\mathbf{m}0}} & \frac{\partial \lambda_v(t_f)}{\partial t_f} \\ \frac{\partial \lambda_m(t_f)}{\partial \lambda_{\mathbf{r}0}} & \frac{\partial \lambda_m(t_f)}{\partial \lambda_{\mathbf{v}0}} & \frac{\partial \lambda_m(t_f)}{\partial \lambda_{\mathbf{m}0}} & \frac{\partial \lambda_m(t_f)}{\partial t_f} \end{bmatrix}_{8 \times 8}$$

From the linearization of the flow it is possible to relate the final and the initial state through the STM:

$$\begin{pmatrix} \delta \mathbf{r}(t_f) \\ \delta \mathbf{v}(t_f) \\ \delta m_f \\ \delta \lambda_r(t_f) \\ \delta \lambda_v(t_f) \\ \delta \lambda_m(t_f) \end{pmatrix}_{14 \times 1} = \begin{bmatrix} \Phi_{\mathbf{r}\mathbf{r}} & \Phi_{\mathbf{r}\mathbf{v}} & \Phi_{\mathbf{r}m} & \Phi_{\mathbf{r}\lambda_r} & \Phi_{\mathbf{r}\lambda_v} & \Phi_{\mathbf{r}\lambda_m} \\ \Phi_{\mathbf{v}\mathbf{r}} & \Phi_{\mathbf{v}\mathbf{v}} & \Phi_{\mathbf{v}m} & \Phi_{\mathbf{v}\lambda_r} & \Phi_{\mathbf{v}\lambda_v} & \Phi_{\mathbf{v}\lambda_m} \\ \Phi_{m\mathbf{r}} & \Phi_{m\mathbf{v}} & \Phi_{mm} & \Phi_{m\lambda_r} & \Phi_{m\lambda_v} & \Phi_{m\lambda_m} \\ \Phi_{\lambda_r\mathbf{r}} & \Phi_{\lambda_r\mathbf{v}} & \Phi_{\lambda_r m} & \Phi_{\lambda_r\lambda_r} & \Phi_{\lambda_r\lambda_v} & \Phi_{\lambda_r\lambda_m} \\ \Phi_{\lambda_v\mathbf{r}} & \Phi_{\lambda_v\mathbf{v}} & \Phi_{\lambda_v m} & \Phi_{\lambda_v\lambda_r} & \Phi_{\lambda_v\lambda_v} & \Phi_{\lambda_v\lambda_m} \\ \Phi_{\lambda_m\mathbf{r}} & \Phi_{\lambda_m\mathbf{v}} & \Phi_{\lambda_m m} & \Phi_{\lambda_m\lambda_r} & \Phi_{\lambda_m\lambda_v} & \Phi_{\lambda_m\lambda_m} \end{bmatrix}_{14 \times 14} \begin{pmatrix} \delta \mathbf{r}_0 \\ \delta \mathbf{v}_0 \\ \delta m_0 \\ \delta \lambda_{r0} \\ \delta \lambda_{v0} \\ \delta \lambda_{m0} \end{pmatrix}_{14 \times 1}$$

The top-left $[7 \times 7]$ block of the derivatives matrix is constructed by substituting the corresponding state and costate State Transition Matrix (STM) components. The final column represents the derivative of the objective function $\mathbf{F}$ with respect to the final time t_f . Notably, the last entry in this column is zero, as dictated by the imposed transversality condition.

The last rows can be retrieved from the dynamics adopted for the PMP:

$$\begin{cases} \dot{\mathbf{x}} = \frac{\partial H}{\partial \mathbf{x}} \\ \dot{\lambda} = -\frac{\partial H}{\partial \lambda} \end{cases} \rightarrow \begin{cases} \frac{\partial H}{\partial \lambda_0} = \frac{\partial H}{\partial \mathbf{x}(t_f)} \frac{\partial \mathbf{x}(t_f)}{\partial \lambda_0} + \frac{\partial H}{\partial \lambda(t_f)} \frac{\partial \lambda(t_f)}{\partial \lambda_0} = -\dot{\lambda}_f^T \Phi_{\mathbf{x}\lambda} + \dot{\mathbf{x}}_f^T \Phi_{\lambda\lambda} \end{cases}$$

where $\mathbf{x}_f$ and λ_f denote the state and costate respectively at the final time t_f .

Hence the derivatives matrix $\frac{\partial \mathbf{F}}{\partial \mathbf{y}}$ becomes:

$$\frac{\partial \mathbf{F}}{\partial \mathbf{y}} = \begin{bmatrix} \Phi_{\mathbf{r}\lambda_r} 3 \times 3 & \Phi_{\mathbf{r}\lambda_v} 3 \times 3 & \Phi_{\mathbf{r}\lambda_m} 3 \times 1 & \dot{\mathbf{r}}_{3 \times 1} \\ \Phi_{\mathbf{v}\lambda_r} 3 \times 3 & \Phi_{\mathbf{v}\lambda_v} 3 \times 3 & \Phi_{\mathbf{v}\lambda_m} 3 \times 1 & \dot{\mathbf{v}}_{3 \times 1} \\ \Phi_{\lambda_m\lambda_r} 1 \times 3 & \Phi_{\lambda_m\lambda_v} 1 \times 3 & \Phi_{\lambda_m\lambda_m} 1 \times 3 & \dot{\lambda}_m \\ -\dot{\lambda}_r^T \Phi_{\mathbf{r}\lambda_r} + \dot{\mathbf{r}}^T \Phi_{\lambda_r\lambda_r} & -\dot{\lambda}_v^T \Phi_{\mathbf{v}\lambda_v} + \dot{\mathbf{v}}^T \Phi_{\lambda_v\lambda_v} & -\dot{\lambda}_m \Phi_{m\lambda_m} + \dot{m} \Phi_{\lambda_m\lambda_m} & 0 \end{bmatrix}_{8 \times 8}$$

D Rotation to ECI frame

An orbit expressed in the rotating frame, with $\mathbf{x} = (x(t), y(t), \dot{x}(t), \dot{y}(t))$, can be rotated in the ECI frame, with $\mathbf{X} = (X(t), Y(t), \dot{X}(t), \dot{Y}(t))$ using the following:

$$\begin{aligned} X(t) &= (x(t) + \mu)\cos(t) - y(t)\sin(t) \\ Y(t) &= (x(t) + \mu)\sin(t) + y(t)\cos(t) \\ \dot{X}(t) &= (\dot{x}(t) - y(t))\cos(t) - (\dot{y}(t) + x(t) + \mu)\sin(t) \\ \dot{Y}(t) &= (\dot{x}(t) - y(t))\sin(t) + (\dot{y}(t) + x(t) + \mu)\cos(t) \end{aligned}$$

The procedure adopted is the one presented in the reference paper (Topputo, 2013)